

CO2xComm: Porewater SO₄/Cl

July 2026 Samples

2026-09-29

Contents

0.1	Run Information	2
0.2	Assess Standard Curves	3
0.3	Assess Check Standards	4
0.4	Assess Blanks	5
0.5	Assess Duplicates	7
0.6	Calculate mmol/L concentrations & salinity, add dilutions	8
0.7	Assess Analytical Spikes	9
0.8	Check if samples within the range of the standard curve	9
0.9	Check to see if samples run match metadata & merge info	10
0.10	Organize Data	10
0.11	SO ₄ - Visualize Data	10
0.12	Cl - Visualize Data	12

##Add Required Packages

0.1 Run Information

```
##### Run information - PLEASE CHANGE
Date_Run = "2026-09-28" #Date that instrument was run
Run_by = "Victoria Ellis" #Instrument user
Script_run_by = "Victoria Ellis" #Code user
sample_year <- "2026"
sample_month <- "July"
Site = "GCRew"
Project_name = "CO2xCOMM"
run_notes = "" #any notes from the run

##### File Names - PLEASE CHANGE
#file path and name for raw summary data file
raw_file_name_cl = "Raw Data/CO2xComm_202607_C1.txt"
raw_file_name_so4 = "Raw Data/CO2xComm_202607_S04.txt"

#file path and name of processed data file
processed_file_name = "Processed Data/GCRew_CO2xCOMM_PW_Processed_C1_S04_202607.csv"

##### Log Files - PLEASE CHECK
Raw_Metadata = here("../", "GCRew", "GCRew_Project_Treatment_Metadata.csv")

#qaqc log file path for this year
Log_path = here("Porewater QAQC Logs", "GCRew_S04_C1_QAQC_Log.csv")
```

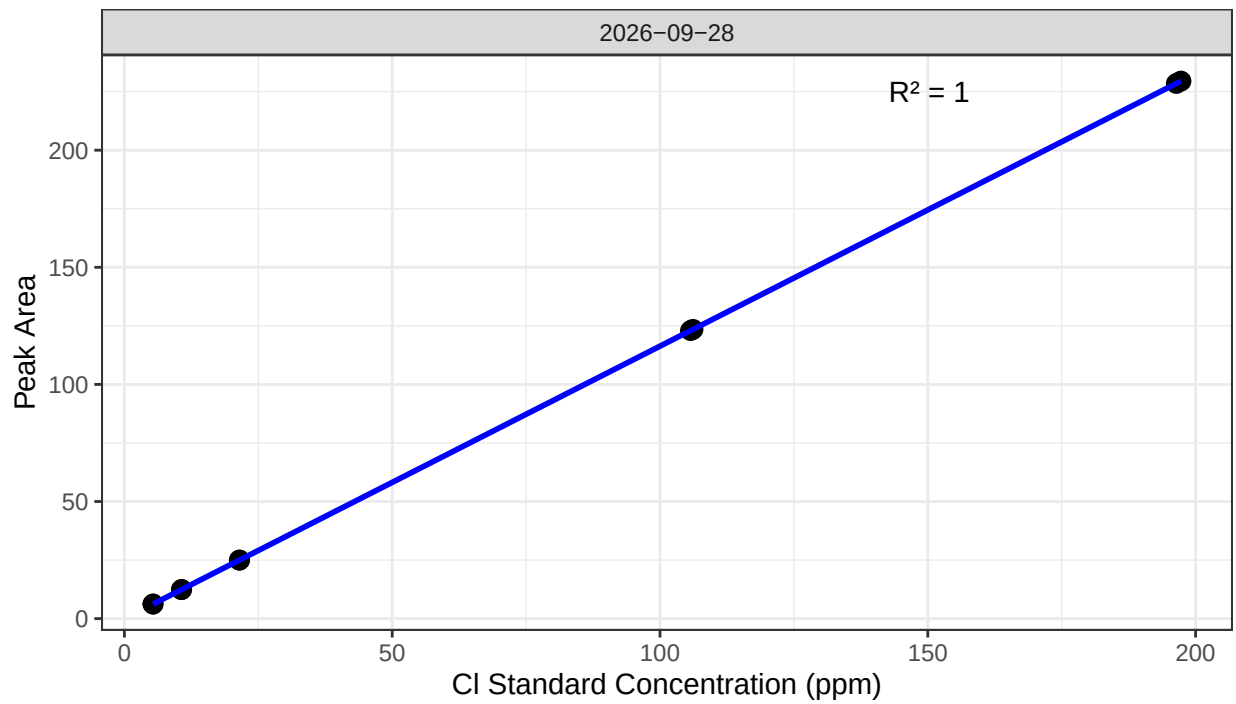
##Set Up Code - constants and QAQC cutoffs

##Read in metadata and create similar sample IDs for matching to samples

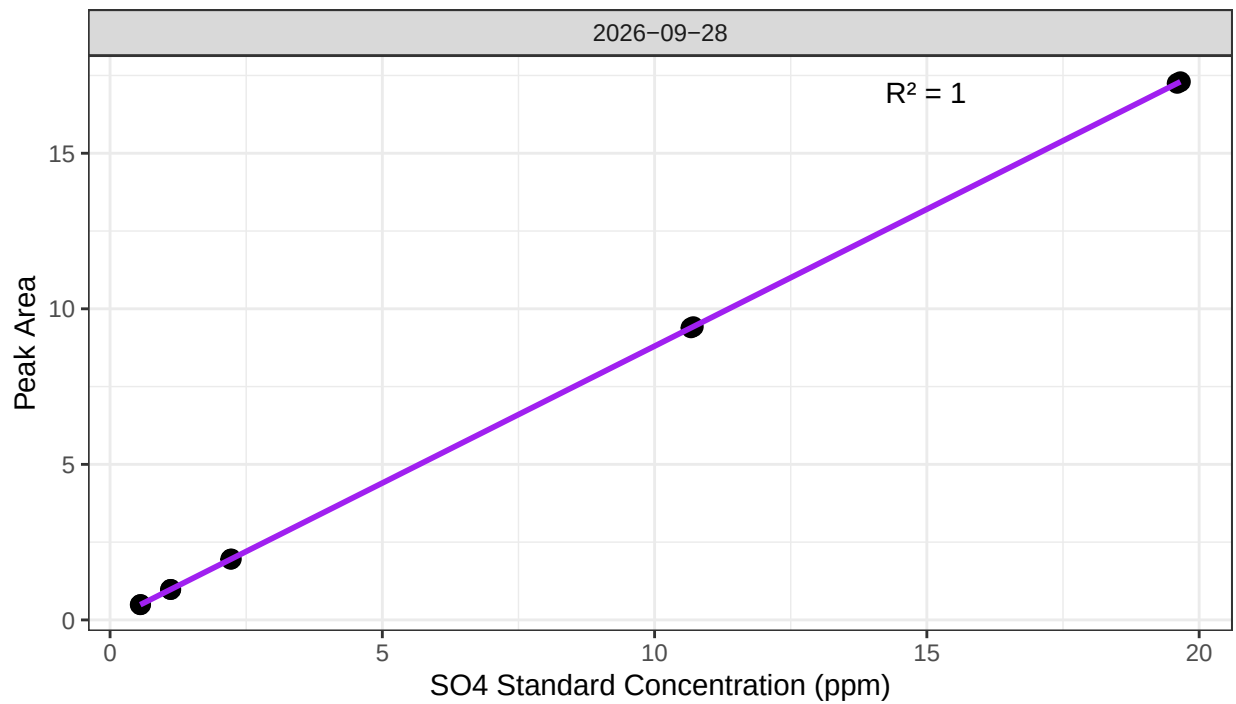
##Import Sample Data

0.2 Assess Standard Curves

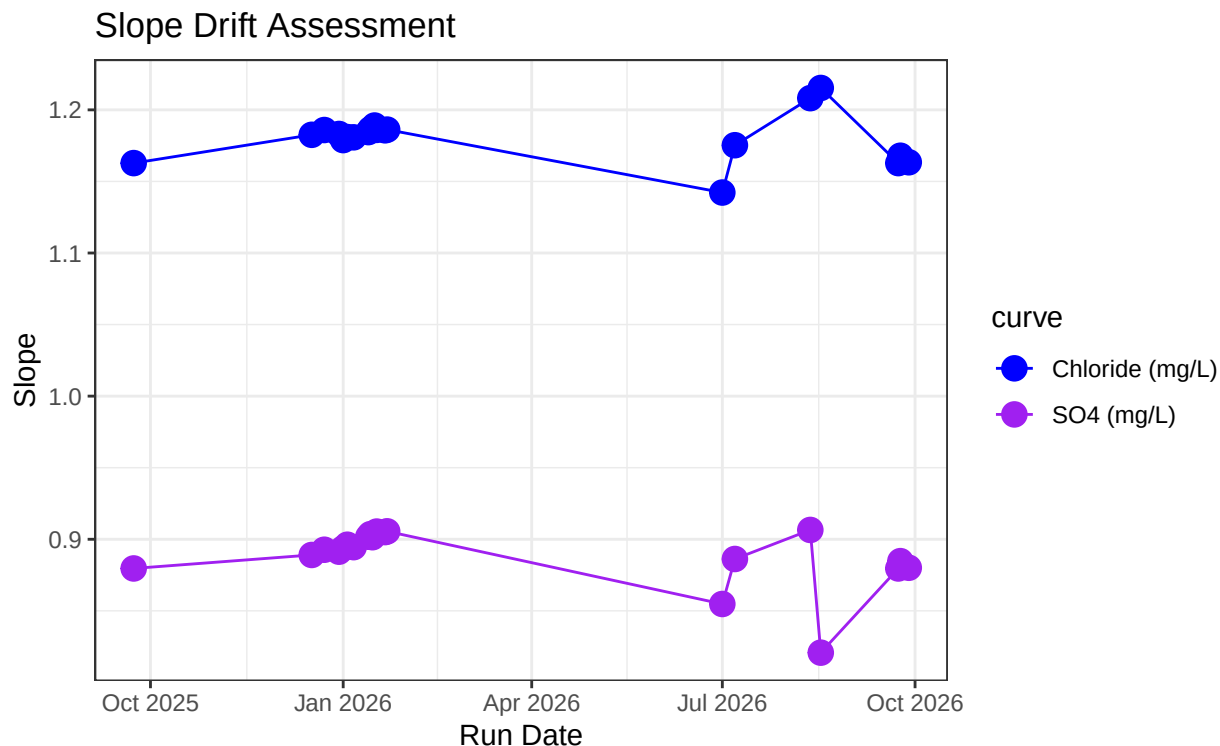
Chloride Std Curve



Sulfate Std Curve



```
## [1] "QAQC log file exists and has been read into the code."
```



```
## [1] "Cl Curve r2 GOOD"
```

```
## [1] "SO4 Curve r2 GOOD"
```

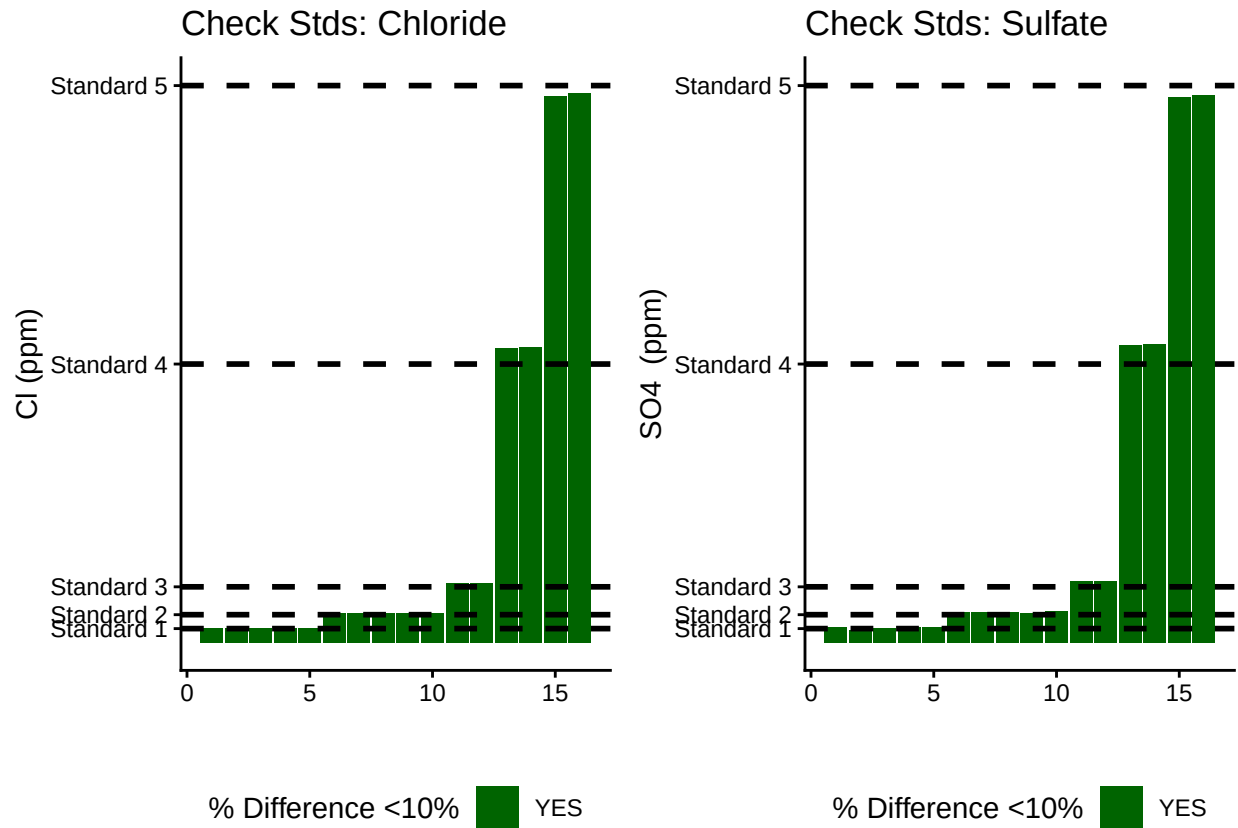
0.3 Assess Check Standards

```
## # A tibble: 5 x 5
##   sample_ID mean_Cl sd_Cl cv_Cl flag_Cl
##   <chr>      <dbl> <dbl> <dbl> <chr>
## 1 Standard 1 5.34 0.0114 0.00213 Chloride Check Standard RSD within Range - ~
## 2 Standard 2 10.7 0.0182 0.00171 Chloride Check Standard RSD within Range - ~
## 3 Standard 3 21.5 0.0389 0.00181 Chloride Check Standard RSD within Range - ~
## 4 Standard 4 106. 0.317 0.00299 Chloride Check Standard RSD within Range - ~
## 5 Standard 5 197. 0.609 0.00309 Chloride Check Standard RSD within Range - ~
```

```
## # A tibble: 5 x 5
##   sample_ID mean_SO4 sd_SO4 cv_SO4 flag_SO4
##   <chr>      <dbl> <dbl> <dbl> <chr>
## 1 Standard 1 0.526 0.0510 0.0970 Sulfate Check Standard RSD within Range --
## 2 Standard 2 1.10 0.0162 0.0147 Sulfate Check Standard RSD within Range --
## 3 Standard 3 2.22 0.00693 0.00312 Sulfate Check Standard RSD within Range --
## 4 Standard 4 10.7 0.0296 0.00277 Sulfate Check Standard RSD within Range --
## 5 Standard 5 19.6 0.0414 0.00211 Sulfate Check Standard RSD within Range --
```

```
## [1] ">80% of Chloride Check Standards have RSD within range - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards have RSD within range - PROCEED"
```



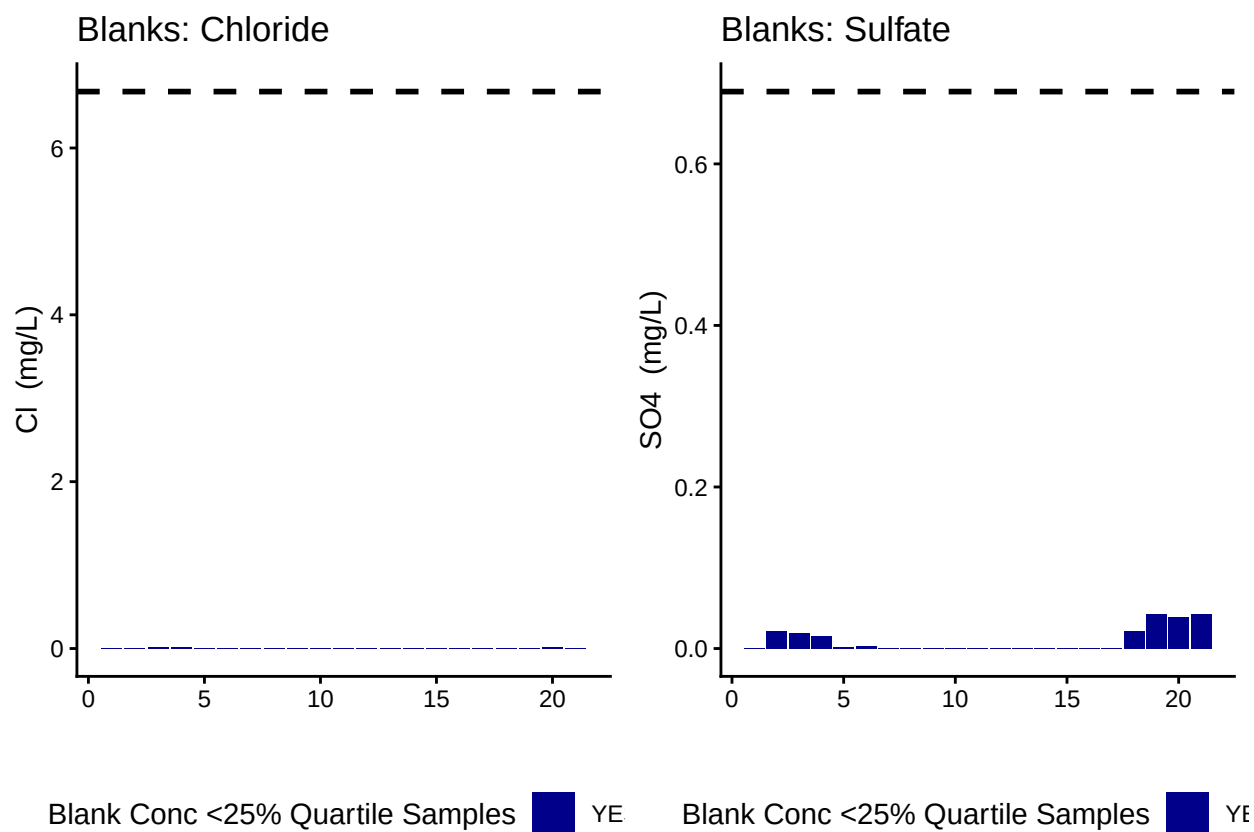
```
## [1] ">80% of Chloride Check Standards are within range of expected concentration - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards are within range of expected concentration - PROCEED"
```

0.4 Assess Blanks

```
## [1] ">80% of Chloride Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">80% of Sulfate Blank concentrations are lower 25% quartile of samples"
```



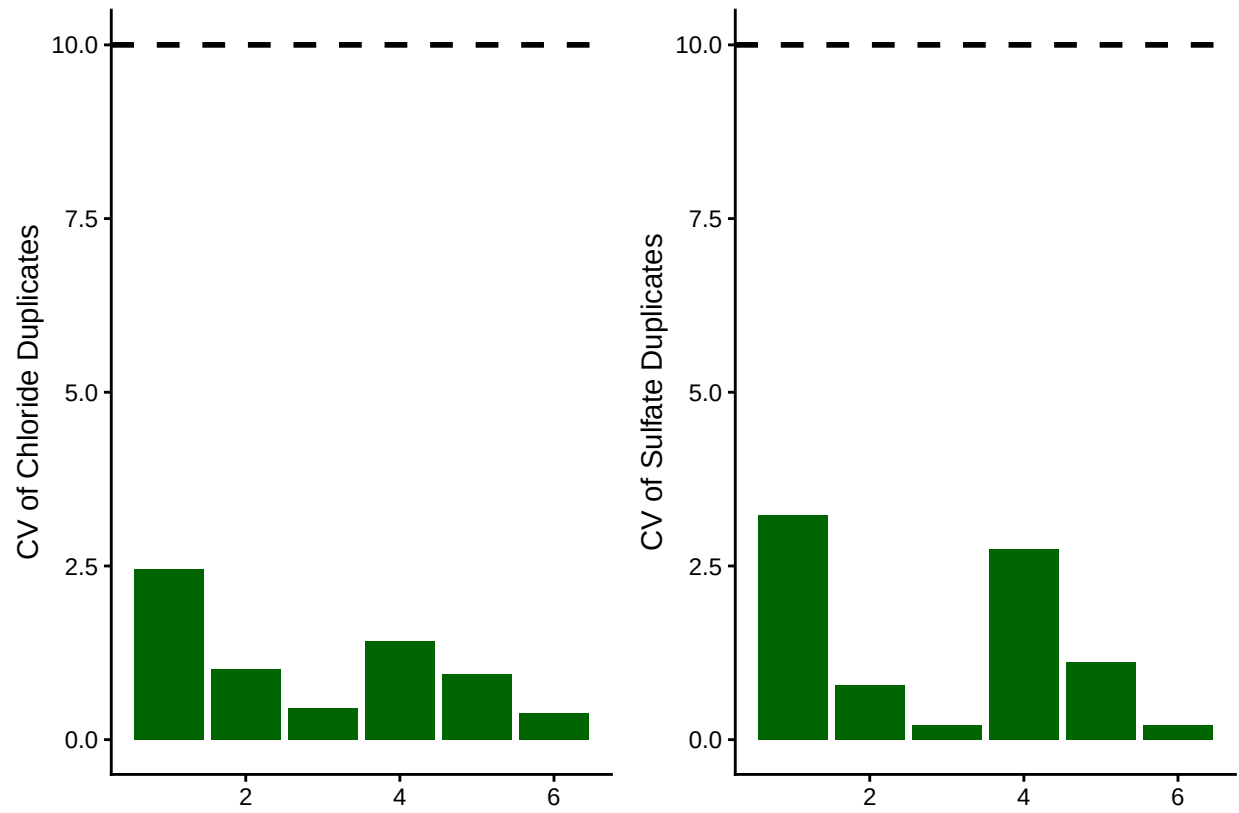
```
## Chloride blanks mean ppm:
```

```
## [1] 0.003104762
```

```
## Sulfate blanks mean ppm:
```

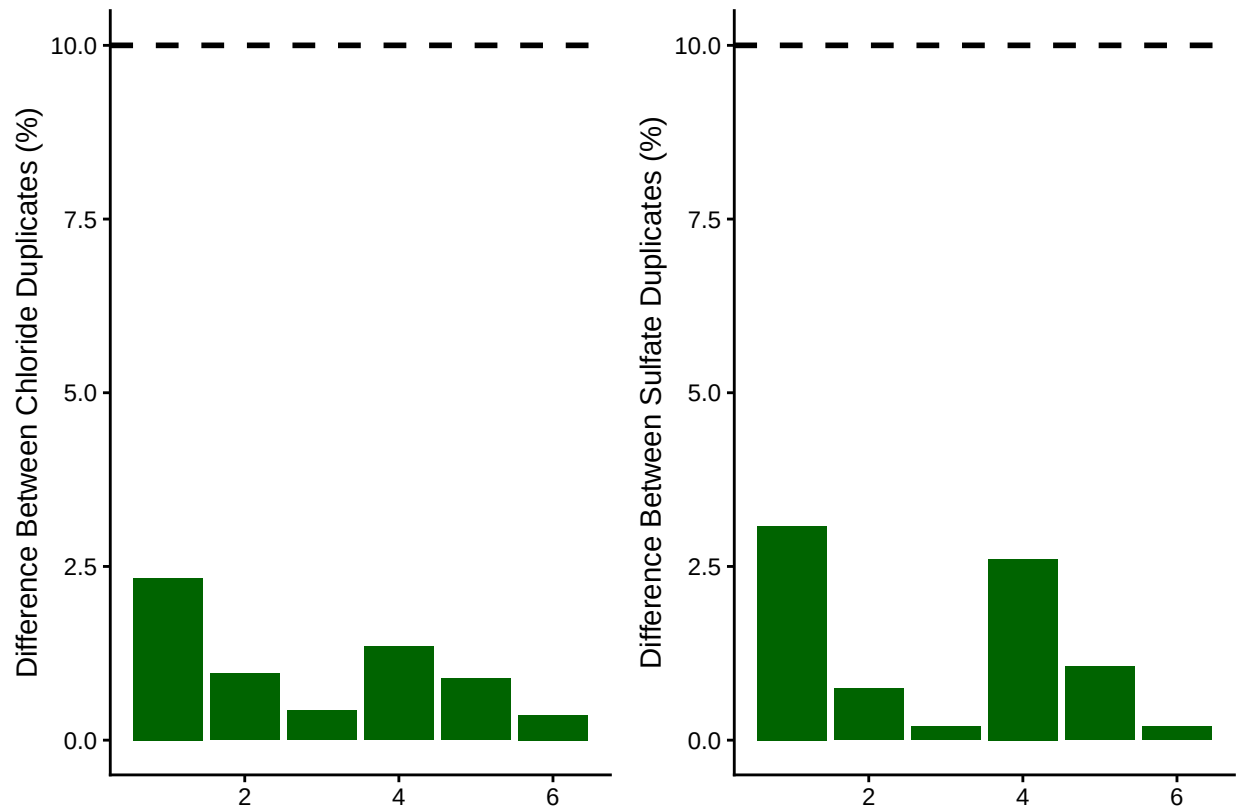
```
## [1] 0.009828571
```

0.5 Assess Duplicates



```
## [1] ">80% of Chloride Duplicates have a CV <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a CV <10% - PROCEED"
```



```
## [1] ">80% of Chloride Duplicates have a percent difference <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a percent difference <10% - PROCEED"
```

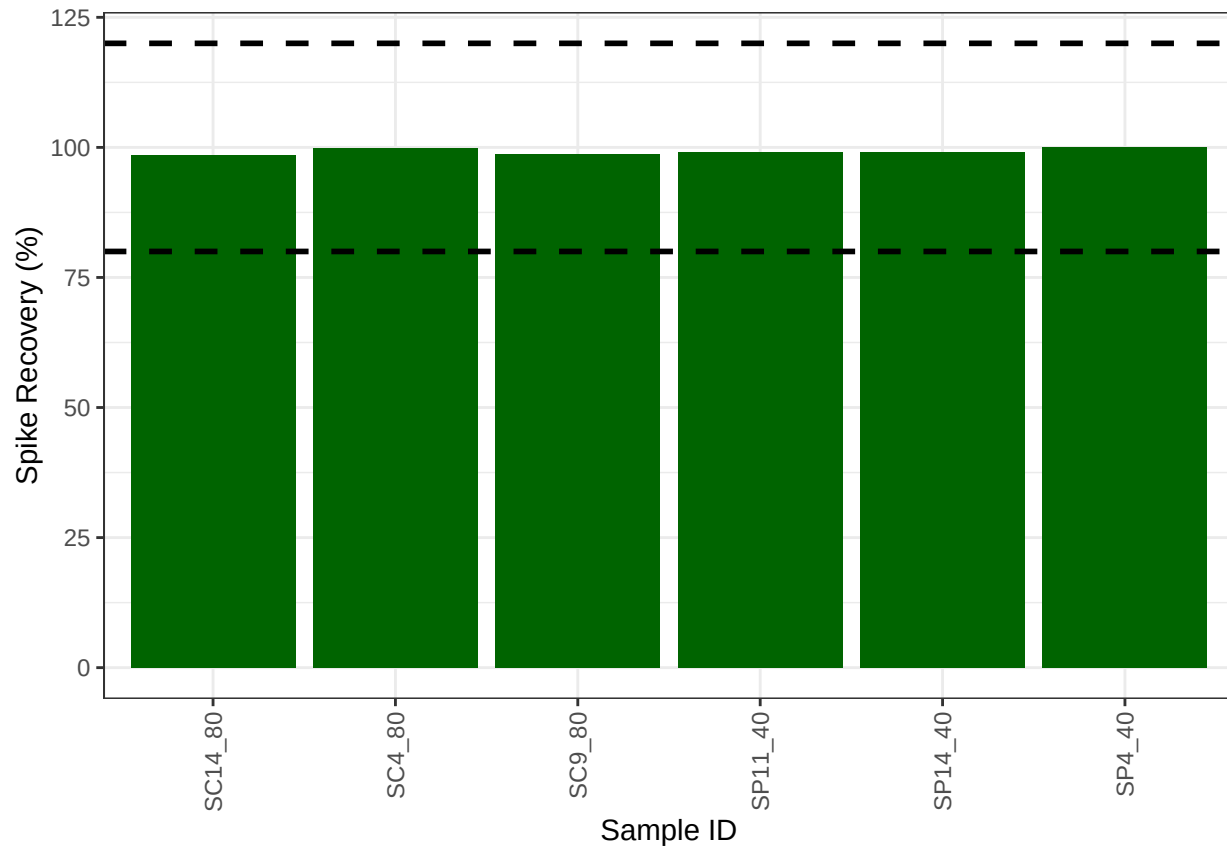
0.6 Calculate mmol/L concentrations & salinity, add dilutions

```
# Convert ppm to mmol/L
all_dat$S04_Conc_mM <- (all_dat$S04_ppm / s_mw)
all_dat$Cl_Conc_mM <- (all_dat$Cl_ppm / cl_mw)

# Calculate Salinity
# calculated using the Knudsen equation
# Salinity = 0.03 + 1.8050 * Chlorinity
# Ref: A Practical Handbook of Seawater Analysis by Strickland & Parsons (P. 11)
# = ((1.807*Cl_ppm)+0.026)/1000
all_dat$salinity <- ((1.8070 * all_dat$Cl_ppm) + 0.026) / 1000

#Need to determine dilution factors for your samples
#for Steph / COMPASS this depends on the site so...
all_dat$Dilution <- Dilution
# head(all_dat)
```


0.7 Assess Analytical Spikes



```
## [1] ">80% of S04 spikes have a recovery between the high and low cutoff - PROCEED"
```

0.8 Check if samples within the range of the standard curve

```
## Sample Flagging
```

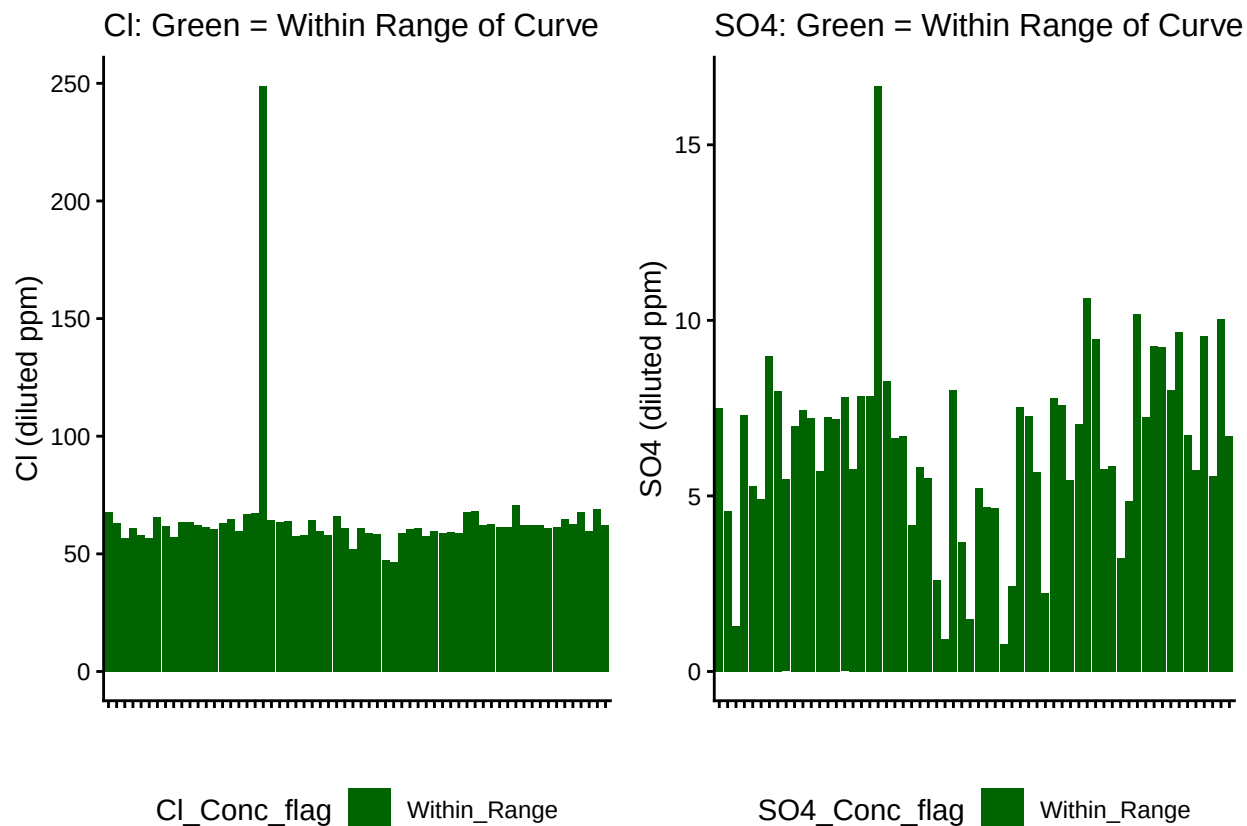


Table 1: SO4 samples

SO4_Conc_flag	Percent_samples
Within_Range	100

Table 2: Cl samples

Cl_Conc_flag	Percent_samples
Within_Range	100

0.9 Check to see if samples run match metadata & merge info

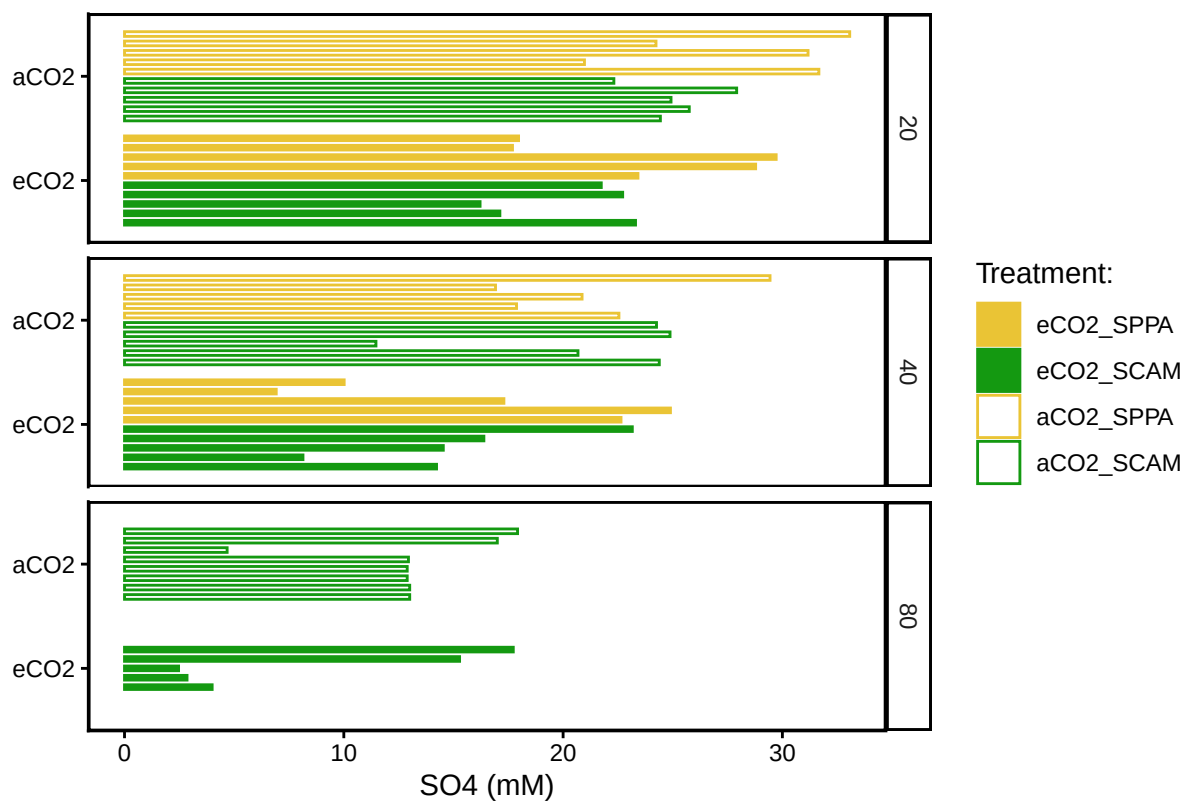
```
## All sample IDs are present in metadata.
```

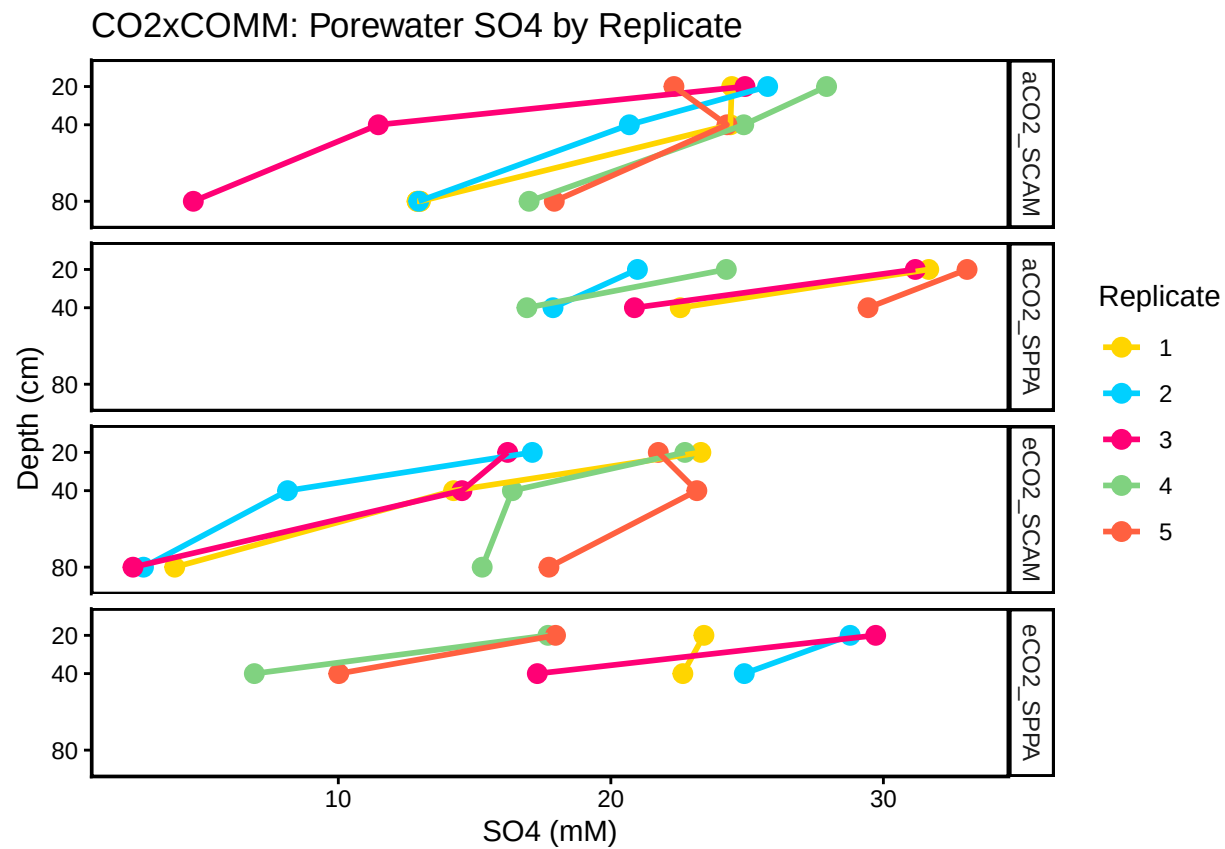
0.10 Organize Data

0.11 SO4 - Visualize Data

```
## Visualize Data
```

CO2xCOMM: Porewater Sulfate

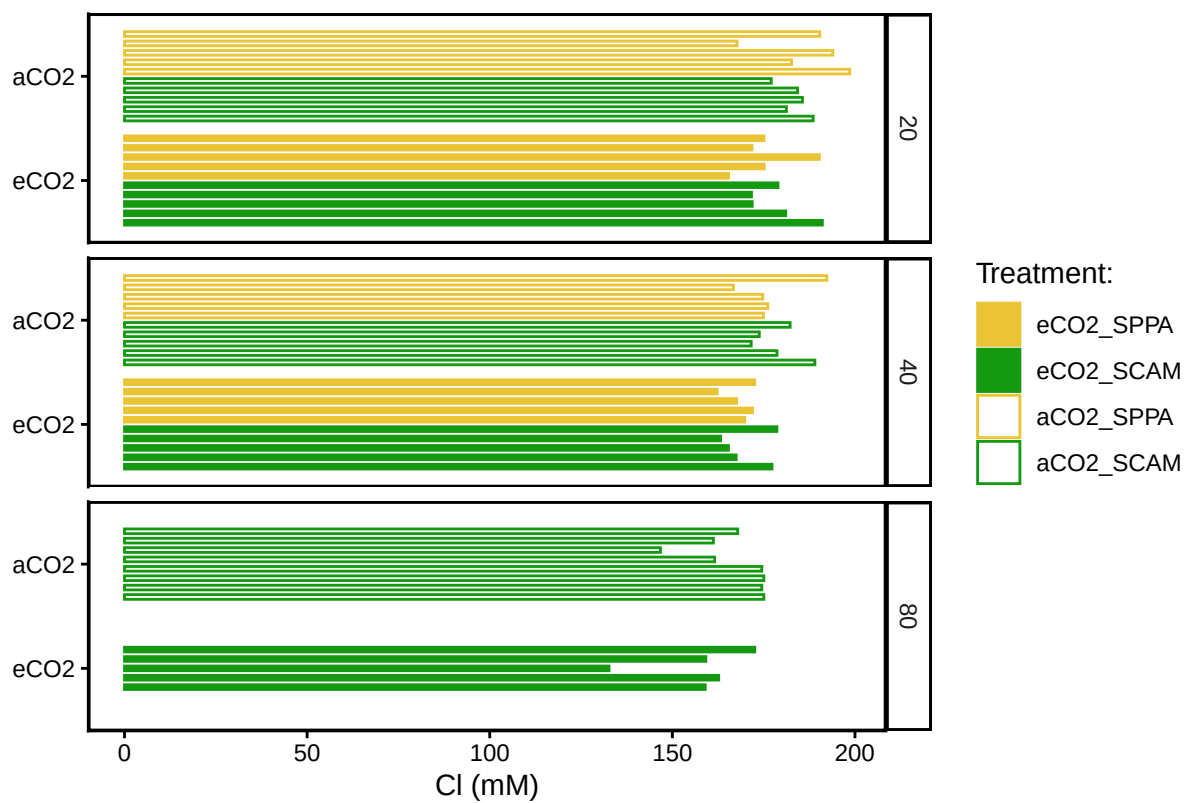


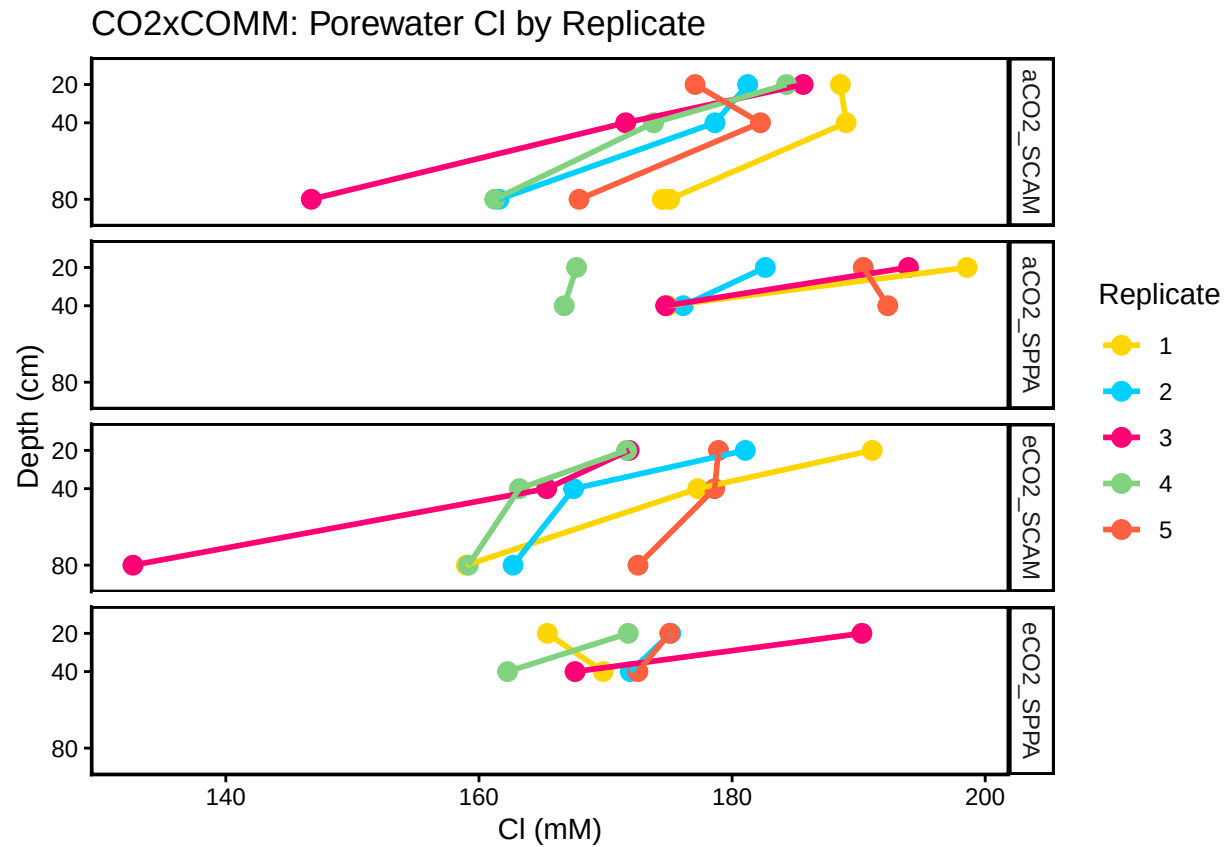


0.12 Cl - Visualize Data

Visualize Data

CO2xCOMM: Porewater Chloride





#Export Data

```
#will put final data in processed data folder
#Write out data frame
write.csv(final_data, processed_file_name)
```

#end